Acoustic Echo Control Based on Sound Object Identification for Suppressing Howling Caused by Complicated Acoustic Paths

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***Abstract*—This paper proposes acoustic echo control based on sound object identification for suppressing acoustic echo and howling in conferencing environments with complicated acoustic paths, where multiple hands-free terminals coexist in the same room. Conventional acoustic echo cancellers target fixed intra-device echo paths; however, unintended paths, for example, those formed via inter-terminal communication, are difficult to control and can lead to howling. Instead of estimating echo paths, the proposed approach identifies sound objects and keeps channels muted by default, allowing pass/playback only when the signal is judged not identical to recently observed objects. This breaks echo loops caused by repeated reproduction of the same sound object and can be viewed as an extension of classical voice switching toward conditional half-duplex operation. Technical challenges and connections to related techniques are discussed, and a simulation shows howling suppression together with a trade-off against speech quality.**

***Index Terms*—Echo Cancellers, Telecommunication, Acoustic Communication, Speech Enhancement.**

I. Introduction

Hands-free terminals such as speakerphones are now routinely used for voice calls. In hands-free communication, acoustic echo and howling due to acoustic feedback from the loudspeaker to the microphone are suppressed by an acoustic echo canceller (AEC) [1], [2], [3], [4], [5]. An AEC estimates the acoustic feedback path with an adaptive filter and cancels the echo. However, when multiple hands-free terminals operate in the same room, unexpected echo paths can arise through a communication server between terminals, and conventional AECs cannot suppress them. Manual mute operations intended to prevent this are unreliable; if several terminals are unmuted at once, echo and howling readily occur. In virtual and metaverse conversations, and with the spread of hearables, intentionally muting devices may become even harder, so solving echo and howling remains important. Proximity-aware multi-device control has begun to appear in practice [6], but it still depends on restricted sensing and inter-device coordination.

This paper proposes acoustic echo control that identifies sound objects and controls their playback, rather than estimating echo paths. Challenges toward realization are discussed, and a simulation shows that howling can be suppressed.

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II. Acoustic Echo and Howling with Multiple Terminals

As a typical difficult case, consider the setup in Fig. 1: one speakerphone-type hands-free terminal A1 and one headset-type terminal A2 placed a short distance apart in the same room. Leakage from the A2 headphones to its microphone is neglected. Terminal A1 includes an AEC (with an echo suppressor), so direct intra-device feedback is removed. The terminals also apply nonlinear processing such as noise suppression (NS) and dynamic range control, and loudspeaker A1 (including its amplifier) exhibits saturation.

Consider the case where the microphones of A1 and A2 are unmuted simultaneously, e.g., to send the A2 talker's voice clearly to another room (Room B in the figure). Sound entering microphone A2 is encoded, sent through the communication server, and played from loudspeaker A1. That playback re-enters microphone A2, forming acoustic echo that can grow into howling.

Usually, unmuting microphone A2 is disallowed, or loudspeaker A1 must be muted first. Such mute discipline breaks conversational flow and is a major reason why voice conferences feel inferior to face-to-face meetings. With hearables, users may join an ongoing call while moving, creating unexpected echo paths.

Conventional path-estimation AECs cannot realistically handle these inter-terminal paths, which include the network, nonlinear in-device processing, delay variation, and user mute actions. Even with deep learning, solving the problem by path estimation remains extremely difficult.

III. Proposed Approach

The proposed approach is called *sound-object-based echo control*. It does not estimate an acoustic path and subtract echo. Instead, it identifies whether sound objects such as utterances or room sounds reappear, and controls pass/playback accordingly. Sound objects are taken to be speech segments of tens to hundreds of milliseconds.

A. Basic Policy

To prevent howling, channels are muted by default and unmuted for pass/playback only when a signal is judged *not identical* to sound objects recently observed at the same terminal. Because only sound objects are used, without path information, echo loops can in principle be broken even for

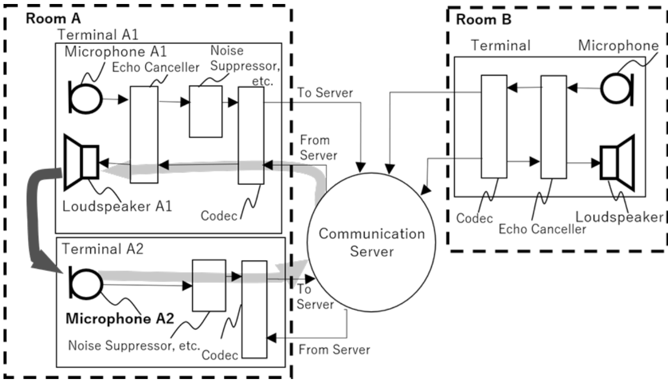

Fig. 1. Echo paths with multiple hands-free terminals in the same room. Each terminal consists of a microphone, AEC, noise suppressor, codec, and loudspeaker, interconnected via a communication server. Leakage from the A2 headphones is neglected. A signal from microphone A2 can be played back from loudspeaker A1 through the server and return to microphone A2, forming a howling loop.

unintended paths that include network delay and nonlinear in-device processing. This control can be viewed as an extension of classical voice-switched half-duplex operation [7], [8] with sound object identification: not permanent half-duplex, but conditional gating that closes locally when identity is detected.

Figure 2 shows a terminal implementation. On the transmit side, a signal is sent to the server only when identification against stored receive-side objects indicates that it is unlikely to stem from the same utterance. On the receive side, playback is allowed only when the receive object is judged different from microphone-side objects. Duplicate reproduction paths of the same object are thereby blocked in principle.

B. Processing Blocks

Processing comprises three elements. End-to-end joint processing may be possible later; here the elements are separated for discussion, challenge analysis, and verification.

1) **Extraction and buffering of sound objects.** Extract objects from the microphone or receive signal and retain them for a duration on the order of the network delay and room reverberation.
2) **Sound object identification.** Compute a similarity or identity probability between the current transmit/receive object and the buffered set.
3) **Playback control.** Control gain from the identification result: mute when identity is likely; pass/playback otherwise. If comparison candidates are insufficient, keep mute (safe-side).

Even if identification errs, a non-persistent error tends to cause only a brief echo rather than sustained howling. Over-muting desired speech, however, reduces intelligibility, so the trade-off between howling suppression and speech quality is fundamental.

IV. Expected Challenges

Extraction and buffering are relatively easy; identification and playback control dominate performance. Placement, coexistence with the existing call pipeline, and training/evaluation design are additional barriers.

A. Identification

Identification must be accurate and low-latency. Extra delay beyond a few tens of milliseconds hurts interactivity, so long-buffer high-accuracy matching is unattractive. Short fragments lack features and raise false-pass errors (same object judged different) and over-muting (different objects judged the same). False-pass errors seed howling; over-muting hurts intelligibility, so the trade-off must be designed explicitly.

Spectra under comparison are strongly deformed from the original: noise and interference, room reverberation, codecs [9], noise/echo suppressors [10], [11], [12], [13], dynamic range control, fragmentation by user mute, sampling-rate and clock mismatch [14], and time warping from packet loss or jitter compensation. Large near/far talker level differences and continuous television or background music in the room also complicate object boundaries and identity decisions. Rephrasing, short backchannels, and repeated music phrases raise identification-error risk even when utterances differ.

Candidate features include Mel-frequency cepstral coefficients [15] and audio fingerprints [16], [17], [18], [19]. Short-segment matching is preferred to limit delay. Fingerprints are robust but need redesign under ultra-low-latency and low-compute call constraints. Longer buffers improve coverage but increase memory, computation, and accidental identification errors. These issues resemble those in source separation and speech enhancement; non-negative matrix factorization and deep learning with deformation-robust representations are promising [20], [21], yet inference delay, on-device compute, and generalization remain constraints.

B. Playback Control

Under simultaneous speech, hard mute is insufficient; separation-like control that retains only needed objects is desirable. Hard mute suppresses howling well but makes identification errors audible as dropouts. Soft gain or frequency-selective mute can ease quality loss, but residual leakage must not re-form a loop. When identification is imperfect, context (e.g., whether audio is from a TV in the same room, or whether the terminal is transmitting) affects the mute decision. Processing order relative to existing AEC, noise suppression, and codecs, and the choice between joint AEC–NS [22] and task splitting [23], are open. Placing the proposed control before or after AEC changes how references are deformed and how hard identification becomes.

C. Placement and System Implementation

Processing may run on terminals, on the server, or both. Terminal-side processing avoids extra uplink bandwidth and favors low delay, but each terminal observes limited signals. With at most one co-located terminal that does not implement the proposed control, howling can be prevented in principle, but loops already formed elsewhere, or loops among unimplemented terminals, cannot be handled. Server-side aggregation

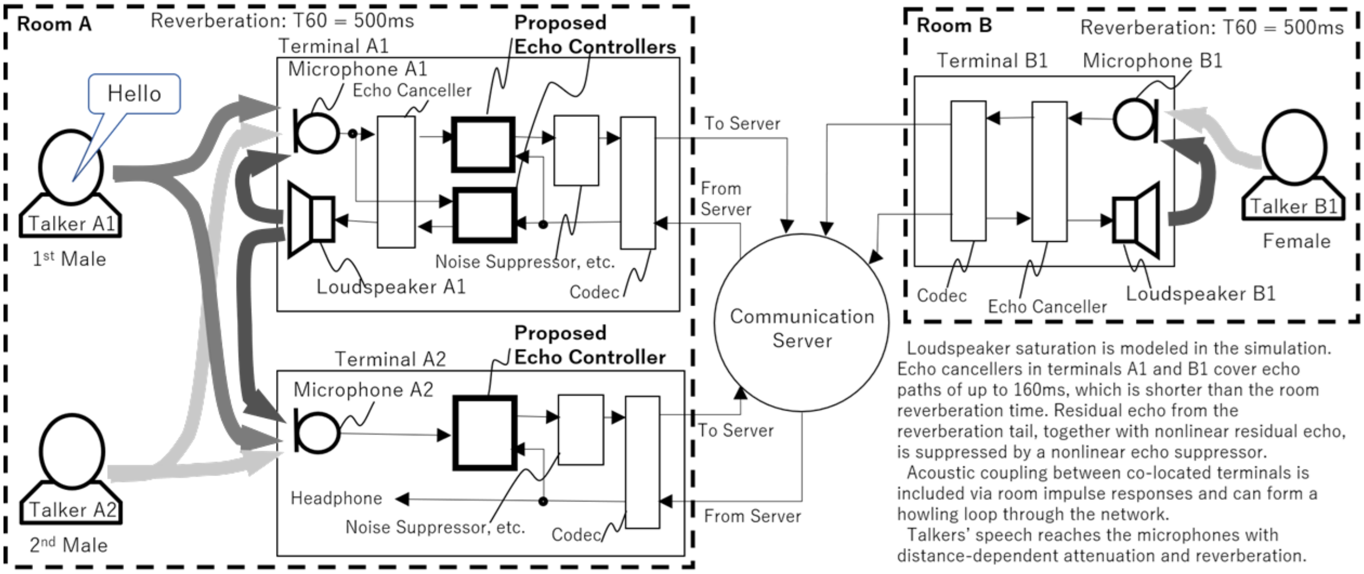

Fig. 2. Proposed echo control and simulation setup. Thick-lined blocks denote the proposed control. On both the microphone and loudspeaker sides, a signal is passed only when sound-object identification indicates that it is likely different from stored objects. The simulation uses two rooms and three terminals: hands-free A1 and microphone-only A2 in Room A (one male talker near each), and hands-free B1 in Room B (one female talker). Both rooms have $T_{60} = 500\,\mathrm{ms}$.

can improve identification but raises uplink delay, privacy, and load. Under partial deployment, safe-side mute may over-mute and make calls impractical. Thresholds and buffer lengths also depend on room, device, and codec conditions, so stable operation without site-specific tuning is difficult.

D. Training and Evaluation

Public data that jointly cover network deformation, in-device nonlinear processing, and playback control are scarce. Conventional AEC benchmarks [24] focus on intra-device echo and do not cover howling via inter-terminal network paths. Subjective quality and intelligibility assessment is also hard [25], [26]; metrics must jointly capture howling suppression and over-mute degradation. For supervised learning, defining identity labels (how much time shift and deformation still count as the same object) is itself a research problem. For privacy, storing fingerprints or embeddings rather than raw waveforms is preferable.

V. Relation to Other Technologies

The proposed approach extends voice-switched half-duplex [7], [8]; near-term use is echo control and howling suppression [27]. As half-duplex evolved into full-duplex via AEC, adaptive processing, source separation, and deep learning can further improve performance.

In using reference signals, it resembles adaptive noise cancellation [28] and crosstalk cancellation [29], and can be seen as extending multi-channel AEC that is cast as source separation [30] to settings that include mute and communication paths. It may also serve as a howling canceller for public-address systems, but repeated sounds such as music raise identification-error risk. Unlike personalized speech enhancement [25], no speaker enrollment is required. End-to-end learning that unifies identification and playback control is conceivable: high-accuracy, delay-tolerant systems can provide data to fine-tune low-latency models.

VI. Verification Simulation

To demonstrate the feasibility of sound-object-based echo control and to illustrate practical issues, a verification simulation is conducted in the two-room, three-terminal setup of Fig. 2. The implementation is an initial gate based on cosine similarity of magnitude spectra, not a final sound-object identifier. Parameters were set empirically so that howling suppression is clear while false-pass errors and over-muting under reverberation and double-talk remain observable.

A. Verification Implementation

Weighted overlap-add analysis is used at 16 kHz with frame length 256 samples (16 ms) and hop 128 samples (8 ms). Utterance onset is declared when frame magnitude-spectrum energy exceeds a threshold; termination follows four hops (32 ms) below threshold, including two preceding hops (16 ms). This simplifies energy-based endpointing [31] with hangover [32]. Completed microphone-side objects are kept for 2.0 s. To limit decision delay, non-overlapping micro-objects of about 96 ms are formed, and candidates from the most recent 0.4 s (0.35 s for transmit-side references) are compared. Receive-side loudspeaker control matches the latest six frames (48 ms) every hop.

Identification uses cosine similarity of magnitude-spectrum sequences [17], [18]. Integer lags of ± 4 hops (± 32 ms) absorb propagation and processing delay; pairs overlapping by fewer than three frames (24 ms) are discarded. On the receive side,

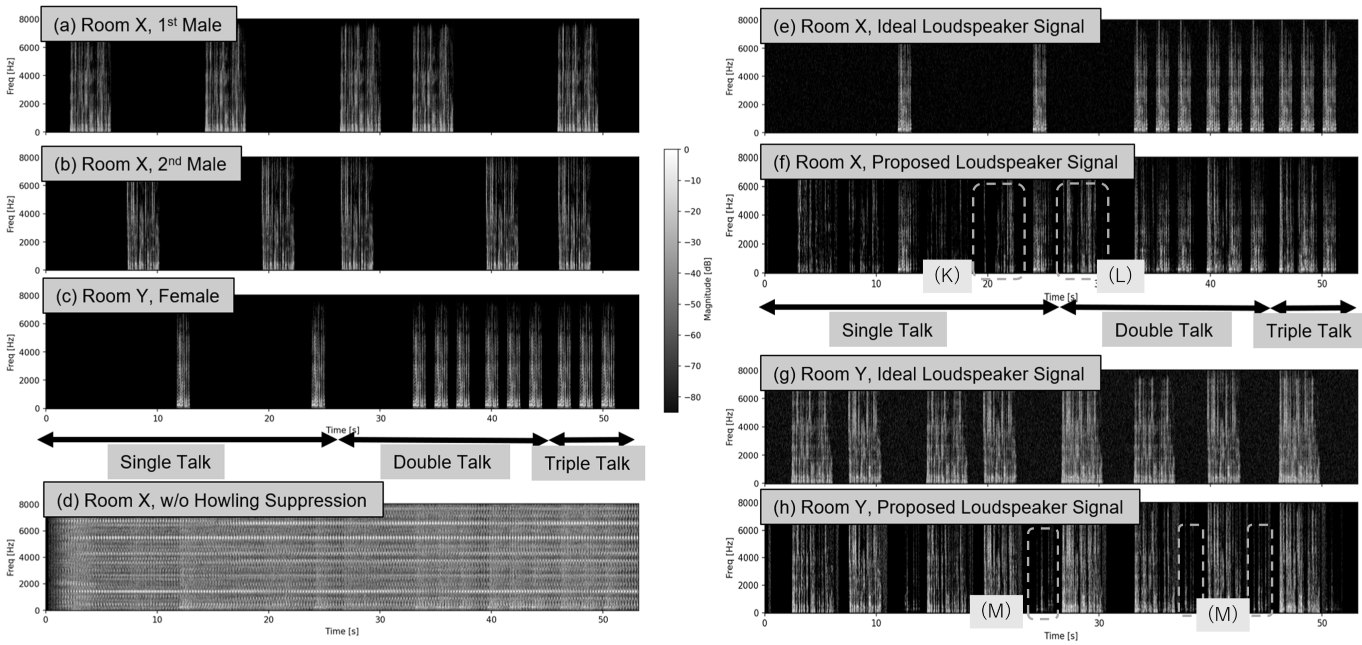

Fig. 3. Spectrograms from the verification simulation. (a)–(c) Talker source signals. (d) Howling without the proposed control. (e),(g) Ideal loudspeaker signals (face-to-face reference) in Rooms A and B. (f),(h) Loudspeaker signals with the proposed control in Rooms A and B. Howling arises readily from terminal coupling in Room A (d), but is suppressed in (f) and (h). Identification errors cause false passes at (K)–(M) and over-muting that thins the spectrograms, indicating that mute-only control is insufficient under double-talk.

pass gain 1.0 is used only if similarity ≤ 0.66; otherwise mute gain 0, also when candidates are missing (safe-side). On the transmit microphone side, mute if similarity ≥ 0.68, else pass. Gains are exponentially smoothed per hop (smoothing factor 0.88). A1 uses both receive mute and transmit mute; A2 uses transmit mute only; automatic mute on B1 is disabled.

B. Call Chain and Talker Scenario

The simulation jointly includes (1) same-room crosstalk, (2) inter-room hands-free loudspeaker-to-microphone feedback, and (3) moderate reverberation ($T_{60} = 500\,\mathrm{ms}$). Background noise is about $-50\,\mathrm{dBFS}$, and the inter-room delay is 200 ms. Each terminal cascades a frequency-domain partitioned AEC [33], [34] with double-talk-robust coefficient smoothing [35], nonlinear residual echo suppression [11], [12], spectral subtraction [36], and an Opus/CELT-like coding distortion model [37].

Talkers are two scheduled male waveforms in Room A and one female in Room B, covering single-talk and double- and triple-talker overlap. Fig. 3(a)–(c) show the sources; (d) shows howling about 2–3 s after call start without the proposed control.

C. Results

Ideal loudspeaker signals as in face-to-face conversation are shown in Fig. 3(e) and (g), and the controlled signals in (f) and (h), respectively. After AEC convergence (about 13 s), single-talk identification errors are relatively few and howling suppression is stable. Double-talk and triple-talk (three-talker overlap) make similarity decisions ambiguous and control harder, yet sustained howling is still suppressed. Momentary echo from identification errors is quickly muted, but over-muting fragments desired speech and reduces intelligibility. The howling-suppression–quality trade-off remains a key topic for improvement.

VII. Conclusion

Sound-object-based echo control is proposed for suppressing howling caused by complicated acoustic paths in multi-terminal voice calls, and the associated challenges are discussed. The proposed approach shifts acoustic echo control from path estimation to sound object identification. Default mute with pass/playback only when non-identity is established breaks unintended echo loops and suppresses howling. In a magnitude-spectrum similarity simulation, howling was suppressed, but speech quality still suffered from over-muting. Improving identification and playback control, especially with deep learning, is future work.

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